

Table 1: Residual-objective ablation on the close-distance frozen test set. Each residual uses identical seed-1337 training frames, fixed backend frame-board inputs, and 3,150 frozen test points. Results are macro-averaged over two disjoint training captures.

<i>(a) Cross-capture image-plane accuracy</i>				
Residual	Train A	Train B	Mean	P95
	Reprojection error [px] ↓			
Pixel	1.0983	1.0206	1.0595	2.0860
Spherical	1.0929	1.0130	1.0529	2.0770
Hybrid	1.0902	1.0104	1.0503	2.0606
<i>(b) View-angle-conditioned ray accuracy</i>				
Residual	0–30°	30–50°	≥ 50°	Ray RMS
	True angular RMSE [mrad] ↓			[deg] ↓
Pixel	0.6079	0.6895	0.8271	0.2100
Spherical	0.6074	0.6857	0.8171	0.1994
Hybrid	0.6068	0.6860	0.8130	0.2044
<i>(c) Tail robustness and optimization cost</i>				
Residual	Overall	Outer	Inlier@1	BA time
	RMSE [px]	RMSE [px]	[%] ↑	[s] ↓
Pixel	1.0595	0.5397	70.17	0.6
Spherical	1.0529	0.5369	70.70	5.5
Hybrid	1.0503	0.5238	70.70	6.4

Protocol. The close-distance test contains no frame used by either training split. Polar bins are assigned with the shared per-training initialization camera; angular errors use the final camera of each method. Ray RMS measures disagreement between the two independently calibrated ray curves over 1,422 image locations.